experiments on human beings, and in January 2024 it was reported that a first brain chip was implanted in a human.

Musk speaks openly about his far-reaching plans for this technology, arguing that it can not only alleviate various medical conditions such as quadriplegia (four-limb paralysis) but also upgrade human abilities and thereby help humankind compete with AI. But it should be clear that at present the Neuralink probes and all other similar biometric devices suffer from a host of technical problems that greatly limit their capabilities. It is difficult to accurately monitor bodily activities – in the brain, heart or anywhere else – from outside the body, whereas implanting electrodes and other monitoring devices into the body is intrusive, dangerous, costly and inefficient. Our immune system, for example, attacks implanted electrodes.[16]

Even more crucially, nobody yet has the biological knowledge necessary to deduce things like precise political opinions from under-the-skin data like brain activity.[17] Scientists are far from understanding the mysteries of the human brain, or even of the mouse brain. Simply mapping every neuron, dendrite and synapse in a mouse brain – let alone understanding the dynamics between them – is currently beyond humanity's computational abilities.[18] Accordingly, while gathering data from inside people's brains is becoming more feasible, using such data to decipher our secrets is far from easy.

One popular conspiracy theory of the early 2020s argues that sinister groups led by billionaires like Elon Musk are already implanting computer chips into our brains in order to monitor and control us. However, this theory focuses our anxieties on the wrong target. We should of course fear the rise of new totalitarian systems, but it is too soon to worry about computer chips implanted in our brains. People should instead worry about the smartphones on which they read these conspiracy theories. Suppose someone wants to know your political views. Your smartphone monitors which news channels you are watching and notes that you watch on average forty minutes of Fox News and forty seconds of CNN a day. Meanwhile, an implanted Neuralink computer chip monitors your heart rate and brain activity throughout the day and notes that your maximum heart rate was 120 beats per minute and that your amygdala is about 5 per cent more active than the human average. Which data would be more useful to guess your political affiliation – the data coming from the smartphone or from the